BORDER GATEWAY PROTOCOL

Cisco CCNP
Period 3-4

By

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Purpose

The goal of this lab is to fully configure internal BGP in IPV4 and IPV6 with RIP, OSPF, and EIGRP connected to BGP. The BGP routing table should have routes to all points of the network and all internal gateway protocols must have BGP routes distributed into them. Three different BGP attributes must be configured in the network.

Background Information

BGP is a unique routing protocol compared to other protocols such as OSPF, EIGRP, and RIP. OSPF, RIP, and EIGRP are all interior gateway protocols, meaning all IGPs are used to forward traffic only to other routers that also run its routing protocol. BGP, however, can take in all of the routes that these protocols distribute and create its own routing table, combining all of these protocols together. A router running OSPF or EIGRP can distribute its routes into a BGP running router, which would allow the BGP network to connect these IGPs together. Similarly, a BGP router can redistribute all its own routes into other IGPs, allowing a router running OSPF to contact all other routers in the topology. This allows BGP to have an incredibly large routing table, allowing it to store the entire topology's routes. This might turn some away from using BGP because of the assumption that a larger routing table might mean a heavy CPU and memory usage. However, BGP dynamically updates its own topology through other routing protocols unlike protocols such as OSPF, which distribute Link State Advertisements. The ability of OSPF to scale is greatly limited because as the network gets larger, more and more Link State advertisements will be broadcast and processed. BGP does not use Link State Advertisements, but rather will instead rely on the routes distributed by other routing protocols to determine its own routing protocol. This allows BGP to store a huge number of prefixes and supply redundancy to routes that are down. For this reason, extremely large-scale networks such as the networks of ISPs and corporate WLANs. Similarly, manipulating BGP attributes such as weight or route aggregation can also allow a simpler and more compressed routing table. As such, while BGP may be extremely scalable, the convergence time of BGP is significantly slower than other protocols such as EIGRP or OSPF. Many BGP attributes can be manipulated by network administrators to influence how routers will determine the path their data will take to the destination. The weight attribute is generally what determines what path packets will take. The higher the weight, the more preferred the path is. Local preference is very similar to the weight attribute, where a higher local preference would mean a more preferred path. The difference between that router's local preference is shared throughout the network. If the weight attributes are the same, then the packets will be sent through the point with the highest local preference. BGP origin code has three flavors: IGP, EGP, and incomplete. IGP means that the BGP path originated from an Interior Gateway Protocol like OSPF, EIGRP, or RIP. EGP indicates that the path originated from an External Gateway protocol, or just BGP. Incomplete indicates that the origin was unable to be determined. IGP is preferred over EGP which is preferred over

incomplete. You can modify this origin code to determine preference of BGP routes. The sixth BGP attribute is the metric, which is used when there are multiple paths into your AS. The metric tells the BGP router's neighbors' which path is preferred for them to enter the BGP network. These attributes may seem complicated but are highly beneficial to learn. Configuring BGP attributes allow for much more flexible modification on the way packets are routed across your network, whether its outbound traffic or how prefixes are exchanged.

Lab Summary

To insert routes into the BGP routing table, we first configured three other networks using OSPF, RIP, and EIGRP. On the interior routers that are connected to both BGP and and an IGP, we made the IGP redistribute all its routes into BGP so any router in the BGP's network is able to ping all other nodes in the network. However, there are no routes on the IGP network routers, so they are unable to respond to the pings as they do not have a path to return to the source ping. Therefore, we then went on all ABRs and redistributed the BGP routes using the redistribute BGP command. This injected all bgp routes into the IGP's routing table. We then configured route aggregation on router 2 by simplifying the routes to the 192.168.0.1/25 and 192.168.0.127/25 into 192.168.0.0/24. Then configured atomic aggregation on router 3, pointing towards router 2. Then configured a weight attribute on

Lab Commands

Clear ip bgp *

Clears all bgp instances and re-initiates bgp adjacency with neighbors.

Show bgp ipv6 unicast

Shows all ipv6 BGP routes including the directly connected ones. Shows suppressed routes.

Show ip bgp

Shows all ipv4 bgp routes. Shows suppressed routes.

Access-list [number] permit [ip-address of the route you aggregated]

Creates an access list that only allows the address you specified to pass.

Route-map (name) permit (access-list number)

Creates a route map and permits the ip address of the access-list specified

Set as-path prepend 650014 650014 650014 650014

Manually sets the as-path which is the preferred route to the destination. A shorter as-path means more preferred.

Router bgp [as number]

Network [ip address] mask [subnet mask]

Advertise this network to all other BGP routers

Neighbor [ip address] remote-as [as-number]

Attempts to create a BGP peer with this ip address

Neighbor [ip address] activate

Establishes BGP adjacency with this ip address

Neighbor [ip address] weight [weight]

Manually changes the weight of the path to the neighbor to a specified value.

Redistribute [IGP] [as-num of the router] metric [metric] include-connected

Will redistribute all the routes received from the routing protocol and inject it into BGP.

Neighbor [ip-address] route-map [name] out

Advertises this route map to a neighbor's address.

Aggregate-address [ip-address] [subnet mask] summary-only

Will aggregate a specified ip address range into one network.

Router [routing protocol] [as-number]

Redistribute bgp [as-number]

redistributes all BGP routes into the routing protocol

Configurations:

R1

```
interface GigabitEthernet0/0/0
ip address 192.168.0.1 255.255.255.128
negotiation auto
ipv6 address 2001:DB8:A::1/64
ipv6 enable
ipv6 ospf 10 area 0
```

```
interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:D::1/64
ipv6 enable

interface Serial0/1/0

interface Serial0/1/1

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router ospf 10
router-id 1.1.1.1
redistribute bgp 1 subnets
network 192.168.0.0 0.0.0.127 area 0
network 192.168.0.0 0.0.0.255 area 0
network 192.168.0.0 0.0.7.255 area 0
```

```
router bgp 1
bgp router-id 1.1.1.1
bgp log-neighbor-changes
neighbor 2001:DB8:D::2 remote-as 2
neighbor 2001:DB8:D::3 remote-as 3
neighbor 192.168.2.2 remote-as 2
neighbor 192.168.2.3 remote-as 3

address-family ipv4
  network 1.1.1.1 mask 255.255.255.255
  network 192.168.2.0
  redistribute ospf 10 metric 1
  neighbor 2001:DB8:D::2 activate
  neighbor 2001:DB8:D::3 activate
  neighbor 192.168.2.2 activate
  neighbor 192.168.2.2 weight 55555
  neighbor 192.168.2.2 route-map map out
  neighbor 192.168.2.3 activate
exit-address-family

address-family ipv6
  redistribute ospf 10 include-connected
  network 2001:DB8:D::/64
  neighbor 2001:DB8:D::2 activate
  neighbor 2001:DB8:D::3 activate
exit-address-family

ip forward-protocol nd
```

```
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0


access-list 10 permit 192.168.2.0 0.0.0.255
ipv6 router ospf 10
router-id 1.1.1.1
redistribute bgp 1


route-map permit permit 10
match ip address 10
set as-path prepend 650014 650014 650014 650014
```

R2

```
router eigrp 2
network 192.168.3.0
redistribute bgp 2 metric 2000 0 255 1 4
eigrp router-id 2.2.2.2

router bgp 2
bgp log-neighbor-changes
neighbor 2001:DB8:D::1 remote-as 1
neighbor 2001:DB8:D::3 remote-as 3
neighbor 192.168.2.1 remote-as 1
neighbor 192.168.2.3 remote-as 3
```

```
address-family ipv4
  network 192.168.2.0
  aggregate-address 192.168.0.0 255.255.255.0 summary-only
  redistribute eigrp 2
  no neighbor 2001:DB8:D::1 activate
  no neighbor 2001:DB8:D::3 activate
  neighbor 192.168.2.1 activate
  neighbor 192.168.2.3 activate
  neighbor 192.168.2.3 weight 1
  neighbor 192.168.2.3 route-map map out
exit-address-family

address-family ipv6
  redistribute eigrp 10 include-connected
  redistribute eigrp 2 metric 1 include-connected
  network 2001:DB8:D::/64
  network FE80::/64
  neighbor 2001:DB8:D::1 activate
  neighbor 2001:DB8:D::3 activate
  neighbor 192.168.2.3 activate
  neighbor 192.168.2.3 route-map map out
exit-address-family

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
access-list 10 permit 192.168.0.0
ipv6 router eigrp 10
eigrp router-id 2.2.2.2
redistribute bgp 2 metric 4 0 255 1 5

ipv6 router eigrp 2
eigrp router-id 22.22.22.22



route-map map permit 10
match ip address 10
set as-path prepend 650014 650014 650014 650014
```

R3

```
interface Loopback0
ip address 1.1.1.1 255.255.255.0
ipv6 address FE80::1 link-local

interface GigabitEthernet0/0/0
ip address 192.168.0.2 255.255.255.128
ip ospf 10 area 0
negotiation auto
ipv6 address 2001:DB8:A::2/64
ipv6 enable
ipv6 ospf 10 area 0

interface GigabitEthernet0/0/1
```

```
ip address 192.168.0.130 255.255.255.128
negotiation auto

interface Serial0/1/0

interface Serial0/1/1

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router ospf 10
router-id 3.3.3.3
network 1.1.1.0 0.0.0.255 area 0
network 192.168.0.128 0.0.0.127 area 0
network 192.168.0.0 0.0.0.255 area 0
network 192.168.0.0 0.0.7.255 area 0

router ospf 1

router bgp 1
bgp log-neighbor-changes
```

```
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 10
router-id 33.33.33.33
```

R4

```
interface Loopback0
ip address 4.4.4.4 255.255.255.255
ipv6 address FE80::1 link-local

interface GigabitEthernet0/0/0
ip address 192.168.3.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:B::2/64
ipv6 enable
ipv6 eigrp 10

interface GigabitEthernet0/0/1
no ip address
shutdown
negotiation auto

interface Serial0/1/0
no ip address
Shutdown
```

```
interface Serial0/1/1
no ip address
Shutdown

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown


router eigrp 2
network 4.4.4.4 0.0.0.0
network 192.168.3.0
eigrp router-id 4.4.4.4

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0


ipv6 router eigrp 10
```

```
eigrp router-id 4.4.4.4
```

R5

```
interface Loopback0
ip address 5.5.5.5 255.255.255.0
ipv6 address FE80:: link-local

interface GigabitEthernet0/0/0
ip address 192.168.4.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:C::1/64
ipv6 enable
ipv6 rip rip enable

interface GigabitEthernet0/0/1
ip address 192.168.2.3 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:D::3/64
ipv6 enable

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

router rip
```

```
redistribute bgp 3 metric 1
network 5.0.0.0
network 192.168.4.0
neighbor 192.168.4.2

router bgp 3
bgp router-id 5.5.5.5
bgp log-neighbor-changes
neighbor 2001:DB8:D::1 remote-as 1
neighbor 2001:DB8:D::2 remote-as 2
neighbor 192.168.2.1 remote-as 1
neighbor 192.168.2.2 remote-as 2

address-family ipv4
  network 5.5.5.5 mask 255.255.255.255
  network 192.168.2.0
  aggregate-address 192.168.0.0 255.255.255.0 summary-only
  redistribute rip
  no neighbor 2001:DB8:D::1 activate
  no neighbor 2001:DB8:D::2 activate
  neighbor 192.168.2.1 activate
  neighbor 192.168.2.1 route-map map out
  neighbor 192.168.2.2 activate
  neighbor 192.168.2.2 weight 55555
  neighbor 192.168.2.2 route-map map out
exit-address-family

address-family ipv6
```

```
  redistribute rip rip include-connected
  neighbor 2001:DB8:D::1 activate
  neighbor 192.168.2.1 activate
  neighbor 192.168.2.2 activate
exit-address-family

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0


access-list 10 permit 192.168.2.0 0.0.0.255
access-list 10 permit 192.168.0.0 0.0.0.255
ipv6 router rip rip
redistribute bgp 3 metric 4

route-map map permit 10
match ip address 10
set ip next-hop 192.168.2.1
set as-path prepend 3 3 3 3
```

R6

```
interface Loopback0
ip address 6.6.6.6 255.255.255.0
ipv6 address FE80::1 link-local
ipv6 enable
```

```
interface GigabitEthernet0/0/0
no ip address
shutdown
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.4.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:C::2/64
ipv6 enable
ipv6 rip rip enable

interface GigabitEthernet0/1/0
no ip address
shutdown
negotiation auto

interface GigabitEthernet0/1/1
no ip address
shutdown
negotiation auto

interface Service-Engine0/2/0
no ip address
shutdown

interface GigabitEthernet0
vrf forwarding Mgmt-intf
```

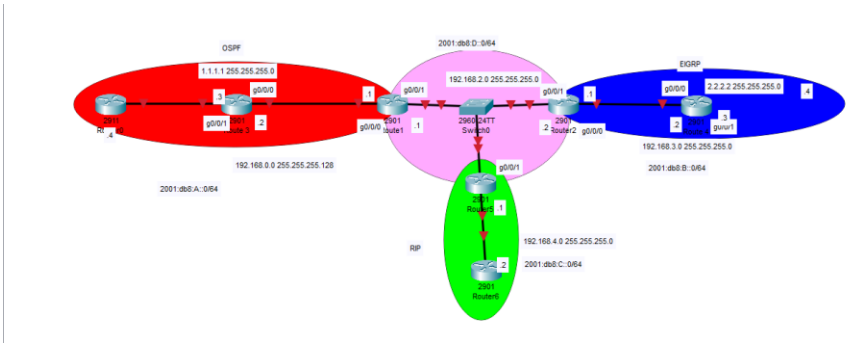
```
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router rip
network 6.0.0.0
network 192.168.4.0
neighbor 192.168.4.1
```

R7

```
interface Loopback0
ip address 8.8.8.8 255.255.255.255

interface GigabitEthernet0/0/0
no ip address
shutdown
negotiation auto

interface GigabitEthernet0/0/1
ip address 192.168.0.129 255.255.255.128
negotiation auto

interface Serial0/1/0
no ip address
```

```
shutdown

interface Serial0/1/1
no ip address
shutdown

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router ospf 10
network 8.8.8.8 0.0.0.0 area 0
network 192.168.0.128 0.0.0.127 area 0
```

Network Diagram + routing tables and pings



R1

IPV4

```
      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 192.168.0.2, 01:18:14,
GigabitEthernet0/0/0
      4.0.0.0/32 is subnetted, 1 subnets
B        4.4.4.4 [20/130816] via 192.168.2.2, 00:33:11
      5.0.0.0/24 is subnetted, 1 subnets
B        5.5.5.0 [20/0] via 192.168.2.3, 00:33:11
B     6.0.0.0/8 [20/0] via 192.168.2.3, 00:33:11
      8.0.0.0/32 is subnetted, 1 subnets
O        8.8.8.8 [110/3] via 192.168.0.2, 01:18:14,
GigabitEthernet0/0/0
      192.168.0.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.0.0/25 is directly connected, GigabitEthernet0/0/0
L        192.168.0.1/32 is directly connected, GigabitEthernet0/0/0
O        192.168.0.128/25
           [110/2] via 192.168.0.2, 01:18:14, GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
B     192.168.3.0/24 [20/0] via 192.168.2.2, 00:33:11
B     192.168.4.0/24 [20/0] via 192.168.2.3, 00:33:11
```

IPV6:

```
C   2001:DB8:A::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:A::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
B   2001:DB8:B::/64 [20/0]
     via FE80::521C:B0FF:FE2D:6801, GigabitEthernet0/0/1
B   2001:DB8:C::/64 [20/0]
```

```
     via FE80::CE7F:76FF:FE6A:ABC1, GigabitEthernet0/0/1
C   2001:DB8:D::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:D::1/128 [0/0]
     via GigabitEthernet0/0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

R2:

IPV4

```
      1.0.0.0/32 is subnetted, 1 subnets
B        1.1.1.1 [20/0] via 192.168.2.1, 00:34:09
      2.0.0.0/32 is subnetted, 1 subnets
C        2.2.2.2 is directly connected, Loopback0
      4.0.0.0/32 is subnetted, 1 subnets
D        4.4.4.4 [90/130816] via 192.168.3.2, 01:19:56,
GigabitEthernet0/0/0
      5.0.0.0/24 is subnetted, 1 subnets
B        5.5.5.0 [20/0] via 192.168.2.3, 00:34:09
B     6.0.0.0/8 [20/1] via 192.168.2.3, 00:34:09
      8.0.0.0/32 is subnetted, 1 subnets
B        8.8.8.8 [20/0] via 192.168.2.1, 00:34:09
B     192.168.0.0/24 [20/0] via 192.168.2.1, 00:34:09
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.2.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.3.1/32 is directly connected, GigabitEthernet0/0/0
B     192.168.4.0/24 [20/0] via 192.168.2.3, 00:34:09
```

IPV6:

```
 B   2001:DB8:A::/64 [20/0]
     via FE80::6EDD:30FF:FEB6:6031, GigabitEthernet0/0/1
C   2001:DB8:B::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:B::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
B   2001:DB8:C::/64 [20/0]
     via 2001:DB8:D::3
C   2001:DB8:D::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:D::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

R3:

IPV4

```
      1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        1.1.1.0/24 is directly connected, Loopback0
L        1.1.1.1/32 is directly connected, Loopback0
      4.0.0.0/32 is subnetted, 1 subnets
O E2     4.4.4.4 [110/1] via 192.168.0.1, 00:34:47,
GigabitEthernet0/0/0
      5.0.0.0/24 is subnetted, 1 subnets
O E2     5.5.5.0 [110/1] via 192.168.0.1, 00:34:47,
GigabitEthernet0/0/0
O E2  6.0.0.0/8 [110/1] via 192.168.0.1, 00:34:47,
GigabitEthernet0/0/0
      8.0.0.0/32 is subnetted, 1 subnets
```

```
O        8.8.8.8 [110/2] via 192.168.0.129, 01:20:50,
GigabitEthernet0/0/1

      192.168.0.0/24 is variably subnetted, 4 subnets, 2 masks

C        192.168.0.0/25 is directly connected,
GigabitEthernet0/0/0

L        192.168.0.2/32 is directly connected,
GigabitEthernet0/0/0

C        192.168.0.128/25 is directly connected,
GigabitEthernet0/0/1

L        192.168.0.130/32 is directly connected,
GigabitEthernet0/0/1

O     192.168.2.0/24 [110/2] via 192.168.0.1, 01:19:50,
GigabitEthernet0/0/0

O E2  192.168.3.0/24 [110/1] via 192.168.0.1, 00:34:47,
GigabitEthernet0/0/0

O E2  192.168.4.0/24 [110/1] via 192.168.0.1, 00:34:47,
GigabitEthernet0/0/0
```

IPV6:

```
C   2001:DB8:A::/64 [0/0]

     via GigabitEthernet0/0/0, directly connected

L   2001:DB8:A::2/128 [0/0]

     via GigabitEthernet0/0/0, receive

OE2 2001:DB8:B::/64 [110/1]

     via FE80::6EDD:30FF:FEB6:6030, GigabitEthernet0/0/0

OE2 2001:DB8:C::/64 [110/1]

     via FE80::6EDD:30FF:FEB6:6030, GigabitEthernet0/0/0

L   FF00::/8 [0/0]

     via Null0, receive
```

R4:

IPV4

```
      1.0.0.0/32 is subnetted, 1 subnets
```

```
D EX     1.1.1.1 [170/1280256] via 192.168.3.1, 00:35:25,
GigabitEthernet0/0/0

      4.0.0.0/32 is subnetted, 1 subnets

C        4.4.4.4 is directly connected, Loopback0

      5.0.0.0/24 is subnetted, 1 subnets

D EX     5.5.5.0 [170/1280256] via 192.168.3.1, 00:35:25,
GigabitEthernet0/0/0

D EX  6.0.0.0/8 [170/1280256] via 192.168.3.1, 00:35:25,
GigabitEthernet0/0/0

      8.0.0.0/32 is subnetted, 1 subnets

D EX     8.8.8.8 [170/1280256] via 192.168.3.1, 00:35:25,
GigabitEthernet0/0/0

D EX  192.168.0.0/24

           [170/1280256] via 192.168.3.1, 00:35:25,
GigabitEthernet0/0/0

D EX  192.168.2.0/24

           [170/1280256] via 192.168.3.1, 01:21:16,
GigabitEthernet0/0/0

      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.3.0/24 is directly connected,
GigabitEthernet0/0/0

L        192.168.3.2/32 is directly connected,
GigabitEthernet0/0/0

D EX  192.168.4.0/24

           [170/1280256] via 192.168.3.1, 00:35:25,
GigabitEthernet0/0/0

Ipv6:

EX  2001:DB8:A::/64 [170/640000256]

     via FE80::521C:B0FF:FE2D:6800, GigabitEthernet0/0/0

C   2001:DB8:B::/64 [0/0]

     via GigabitEthernet0/0/0, directly connected

L   2001:DB8:B::2/128 [0/0]
```

```
     via GigabitEthernet0/0/0, receive
EX  2001:DB8:C::/64 [170/640000256]
     via FE80::521C:B0FF:FE2D:6800, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
```

R5:

IPV4

```
      1.0.0.0/32 is subnetted, 1 subnets
B        1.1.1.1 [20/2] via 192.168.2.1, 00:40:24
      4.0.0.0/32 is subnetted, 1 subnets
B        4.4.4.4 [20/0] via 192.168.2.2, 00:35:55
      5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        5.5.5.0/24 is directly connected, Loopback0
L        5.5.5.5/32 is directly connected, Loopback0
R     6.0.0.0/8 [120/1] via 192.168.4.2, 00:00:05,
GigabitEthernet0/0/0
      8.0.0.0/32 is subnetted, 1 subnets
B        8.8.8.8 [20/1] via 192.168.2.1, 00:40:24
      192.168.0.0/24 is variably subnetted, 3 subnets, 2 masks
B        192.168.0.0/24 [200/0], 00:39:59, Null0
B        192.168.0.0/25 [20/0] via 192.168.2.1, 00:39:59
B        192.168.0.128/25 [20/1] via 192.168.2.1, 00:39:59
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.2.3/32 is directly connected,
GigabitEthernet0/0/1
B     192.168.3.0/24 [20/0] via 192.168.2.2, 00:35:55
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C        192.168.4.0/24 is directly connected,
GigabitEthernet0/0/0

L        192.168.4.1/32 is directly connected,
GigabitEthernet0/0/0
```

IPV6:

```
B   2001:DB8:A::/64 [20/0]

     via FE80::6EDD:30FF:FEB6:6031, GigabitEthernet0/0/1

B   2001:DB8:B::/64 [20/0]

     via FE80::521C:B0FF:FE2D:6801, GigabitEthernet0/0/1

C   2001:DB8:C::/64 [0/0]

     via GigabitEthernet0/0/0, directly connected

L   2001:DB8:C::1/128 [0/0]

     via GigabitEthernet0/0/0, receive

C   2001:DB8:D::/64 [0/0]

     via GigabitEthernet0/0/1, directly connected

L   2001:DB8:D::3/128 [0/0]

     via GigabitEthernet0/0/1, receive

L   FF00::/8 [0/0]

     via Null0, receive
```

R6:

IPV4

```
R     1.0.0.0/8 [120/1] via 192.168.4.1, 00:00:26,
GigabitEthernet0/0/1

R     4.0.0.0/8 [120/1] via 192.168.4.1, 00:00:26,
GigabitEthernet0/0/1

R     5.0.0.0/8 [120/1] via 192.168.4.1, 00:00:26,
GigabitEthernet0/0/1

      6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```

```
C        6.6.6.0/24 is directly connected, Loopback0

L        6.6.6.6/32 is directly connected, Loopback0

R     8.0.0.0/8 [120/1] via 192.168.4.1, 00:00:26,
GigabitEthernet0/0/1

R     192.168.0.0/24 [120/1] via 192.168.4.1, 00:00:26,
GigabitEthernet0/0/1

R     192.168.2.0/24 [120/1] via 192.168.4.1, 00:00:26,
GigabitEthernet0/0/1

R     192.168.3.0/24 [120/1] via 192.168.4.1, 00:00:26,
GigabitEthernet0/0/1

      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.4.0/24 is directly connected,
GigabitEthernet0/0/1

L        192.168.4.2/32 is directly connected,
GigabitEthernet0/0/1
```

IPV6:

```
R   2001:DB8:A::/64 [120/5]

     via FE80::CE7F:76FF:FE6A:ABC0, GigabitEthernet0/0/1

R   2001:DB8:B::/64 [120/5]

     via FE80::CE7F:76FF:FE6A:ABC0, GigabitEthernet0/0/1

C   2001:DB8:C::/64 [0/0]

     via GigabitEthernet0/0/1, directly connected

L   2001:DB8:C::2/128 [0/0]

     via GigabitEthernet0/0/1, receive

L   FF00::/8 [0/0]

     via Null0, receive
```

Pings:

From R3 to all end points

```
R3#ping 192.168.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R3#ping 192.168.4.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R3#ping 2001:db8::c::2
% Unrecognized host or address, or protocol not running.

R3#ping 2001:db8::c::2
% Unrecognized host or address, or protocol not running.

R3#
R3#
R3#ping 2001:db8:b::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:B::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R3#ping 2001:db8:c::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:C::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

```
R3#traceroute 192.168.3.2
Type escape sequence to abort.
Tracing the route to 192.168.3.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.1 0 msec 1 msec 0 msec
  2 192.168.2.2 1 msec 1 msec 1 msec
  3 192.168.3.2 1 msec 1 msec *
R3#
R3#
R3#traceroute 192.168.4.2
Type escape sequence to abort.
Tracing the route to 192.168.4.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.0.1 0 msec 1 msec 1 msec
  2 192.168.2.3 0 msec 2 msec 0 msec
  3 192.168.4.2 1 msec 1 msec *
R3#
R3#
R3#traceroute 2001:db8:b::2
Type escape sequence to abort.
Tracing the route to 2001:DB8:B::2

  1 2001:DB8:A::1 1 msec 1 msec 0 msec
  2 2001:DB8:D::2 1 msec 1 msec 2 msec
  3 2001:DB8:B::2 1 msec 1 msec 1 msec
R3#traceroute 2001:db8:c::2
Type escape sequence to abort.
Tracing the route to 2001:DB8:C::2

  1 2001:DB8:A::1 0 msec 1 msec 1 msec
  2 2001:DB8:D::3 1 msec 1 msec 1 msec
  3 2001:DB8:C::2 1 msec 1 msec 1 msec
```

From R4 to all end points

```
R4>en
R4#ping 192.168.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R4#ping 192.168.4.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R4#ping 2001:db8:a::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:A::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R4#ping 2001:db8:c::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:C::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

```
R4>en
R4#traceroute 192.168.0.2
Type escape sequence to abort.
Tracing the route to 192.168.0.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.3.1 1 msec 0 msec 0 msec
  2 192.168.2.1 0 msec 1 msec 0 msec
  3 192.168.0.2 1 msec 1 msec *
R4#traceroute 192.168.4.2
Type escape sequence to abort.
Tracing the route to 192.168.4.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.3.1 0 msec 2 msec 3 msec
  2 192.168.2.3 2 msec 1 msec 1 msec
  3 192.168.4.2 1 msec 1 msec *
R4#
R4#traceroute 2001:db8:a::2
Type escape sequence to abort.
Tracing the route to 2001:DB8:A::2

  1 2001:DB8:B::1 1 msec 1 msec 1 msec
  2 2001:DB8:D::1 1 msec 1 msec 1 msec
  3 2001:DB8:A::2 1 msec 2 msec 0 msec
R4#traceroute 2001:db8:c::2
Type escape sequence to abort.
Tracing the route to 2001:DB8:C::2

  1 2001:DB8:B::1 1 msec 2 msec 1 msec
  2 2001:DB8:D::3 1 msec 1 msec 1 msec
  3 2001:DB8:C::2 1 msec 1 msec 0 msec
```

From R6 to all endpoints:

```
R6#ping 2001:db8:a::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:A::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R6#ping 2001:db8:b::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:B::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R6#ping 192.168.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R6#ping 192.168.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R6#
```

```
R6#traceroute 192.168.0.2
Type escape sequence to abort.
Tracing the route to 192.168.0.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.4.1 2 msec 1 msec 1 msec
  2 192.168.2.1 2 msec 1 msec 1 msec
  3 192.168.0.2 1 msec 1 msec *
R6#tracert 192.168.2.2
         ^
% Invalid input detected at '^' marker.

R6#traceroute 192.168.2.2
Type escape sequence to abort.
Tracing the route to 192.168.2.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.4.1 1 msec 1 msec 1 msec
  2 192.168.2.2 1 msec 2 msec *
R6#traceroute 2001:db8:a::2
Type escape sequence to abort.
Tracing the route to 2001:DB8:A::2

  1 2001:DB8:C::1 1 msec 1 msec 1 msec
  2 2001:DB8:D::1 1 msec 2 msec 1 msec
  3 2001:DB8:A::2 1 msec 1 msec 1 msec
R6#tracert 2001:db8:b::2
         ^
% Invalid input detected at '^' marker.

R6#traceroute 2001:db8:b::2
Type escape sequence to abort.
Tracing the route to 2001:DB8:B::2

  1 2001:DB8:C::1 1 msec 1 msec 1 msec
  2 2001:DB8:D::2 1 msec 1 msec 1 msec
  3 2001:DB8:B::2 1 msec 1 msec 1 msec
```

BGP Attributes:

```
     Network          Next Hop            Metric LocPrf Weight Path
 *>  1.1.1.1/32       192.168.0.2              2         32768 i
 *   4.4.4.4/32       192.168.2.2                            0 3 2 ?
 *>                   192.168.2.2         130816         11111 2 ?
 *>  5.5.5.0/24       192.168.2.3              0             0 3 ?
 *>  8.8.8.8/32       192.168.0.2              1         32768 ?
 s>  192.168.0.0/25   0.0.0.0                  0         32768 ?
 *>  192.168.0.0      0.0.0.0                            32768 i
 *                   192.168.2.2                            0 3 2 i
 *                   192.168.2.2              0         11111 2 i
 s>  192.168.0.128/25 192.168.0.2              1         32768 ?
 *   192.168.2.0      192.168.2.2              0         11111 2 i
 *                   192.168.2.3              0             0 3 i
 *>                  0.0.0.0                  0         32768 i
 *   192.168.3.0      192.168.2.2                            0 3 2 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>                   192.168.2.2              0         11111 2 ?
 *>  192.168.4.0      192.168.2.3              0             0 3 ?
R1(config-router-af)#
R1(config-router-af)#do sh ip bgp 192.168.0.0
BGP routing table entry for 192.168.0.0/25, version 13
Paths: (1 available, best #1, table default, Advertisements suppressed by an aggregate.)
  Not advertised to any peer
  Refresh Epoch 1
  Local
    0.0.0.0 from 0.0.0.0 (1.1.1.1)
      Origin incomplete, metric 0, localpref 100, weight 32768, valid, sourced, best
      rx pathid: 0, tx pathid: 0x0
```

Normal Route aggregation. Suppressed routes are shown with a S> and when specifically shown with show ip bgp (ip address), the routes are specifically stated to be aggregated. Route aggregation will simplify multiple specific routes into one route than encompases them all.

```
  Advertised to update-groups:
     2
  Refresh Epoch 1
  3 1, (aggregated by 1 1.1.1.1)
    192.168.2.1 from 192.168.2.3 (5.5.5.5)
      Origin IGP, localpref 100, valid, external, atomic-aggregate
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 1
  1, (aggregated by 1 1.1.1.1)
    192.168.2.1 from 192.168.2.1 (1.1.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, atomic-aggregate, best
      rx pathid: 0, tx pathid: 0x0
```

Atomic aggregation is configured on this router. It points towards 192.168.2.1, which is the router stated above and has normal aggregation configured on it. This router will send all traffic that is destined to the 192.168.0.0/24 network to 192.168.2.2/24 because that router has 192.168.0.0/24 aggregated.

```
     Network          Next Hop            Metric LocPrf Weight Path
 *>  1.1.1.1/32       192.168.0.2              2         32768 i
 *   4.4.4.4/32       192.168.2.2                            0 3 2 ?
 *>                   192.168.2.2         130816         11111 2 ?
 *>  5.5.5.0/24       192.168.2.3              0             0 3 ?
 *>  8.8.8.8/32       192.168.0.2              1         32768 ?
 s>  192.168.0.0/25   0.0.0.0                  0         32768 ?
 *>  192.168.0.0      0.0.0.0                            32768 i
 *                   192.168.2.2                            0 3 2 i
 *                   192.168.2.2              0         11111 2 i
 s>  192.168.0.128/25 192.168.0.2              1         32768 ?
 *   192.168.2.0      192.168.2.2              0         11111 2 i
 *                   192.168.2.3              0             0 3 i
 *>                  0.0.0.0                  0         32768 i
 *   192.168.3.0      192.168.2.2                            0 3 2 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>                  192.168.2.2              0         11111 2 ?
 *>  192.168.4.0      192.168.2.3              0             0 3 ?
R1(config-router-af)#
```

Weight attribute. This attribute is used to manipulate which path the packets will take through the network. The higher the weight, the more priority it has.

Problems:

Did not redistribute routes into the IGP areas.

> When we received all the BGP routes in the interior BPG network, we assumed we should have automatically been able to contact all other routers with a
>
> response ping. However, we did not account that the end points were unable to return pings because they themselves did not have a route back to these BGP routers.

Unclear on rip routing

> We were unfamiliar with routing in RIP and made quite a few configuration
>
> mistakes such as not enabling rip routing on the interface and incorrect ipv6 rip routing names.

Unable to configure BGP on packet tracer

> Since we were unable to configure BGP on packet tracer, we had to physically set up the topology anytime we wanted to try and configure BGP, leading to a lot of time spend on the booting process.

Problems with copying a config into the router

> When we pasted BGP configurations into a router, we found that sometimes routers would have a show run that said that BGP neighbor activate commands were active. However, these commands would not actually have been inserted. This meant that we had to re-type in the neighbor activate commands every time we configured the routers, which led to a lot of confusion early on.

Issues with BGP commands

The ipv6 bgp commands and show commands are different from most other IGP commands. An example can be the show bgp ipv6 unicast, whilst in ospf it would be show ipv6 ospf route.

Normal aggregation vs atomic aggregation

Confusion in configuring normal aggregation vs atomic aggregation. When we managed to configure normal aggregation, it was very confusing on how to set up atomic aggregation because it seemed like our neighbors used the same commands and configured atomic aggregation instead.

Conclusion

BGP is a very versatile and scalable routing protocol for tying together different IGP networks to allow connectivity between them whilst also ensuring that routers that run BGP do not consume too much memory and CPU along with the routers that run other routing protocols also do not have to use as much memory to maintain their own linkstate databases. Overall, while BGP is not the most efficient protocol for convergence, BGP is incredibly useful in large scale networks that require scalability over all other aspects.

BGP Signoff

Cornelius Wise P3-4 Cisco CCNP Mr.Mason

Cisco

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